

Guanheng Chen

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EDUCATION

Tsinghua University

Institute for Interdisciplinary Information Sciences (IIIS)

Third-year undergraduate

Beijing, China

Aug 2024 – Present

RESEARCH INTERESTS

Large language model post-training, particularly learning from experience and natural-language feedback.

RESEARCH PAPERS

- [1] **Guanheng Chen***, Tianzhu Ye*, Li Dong*, Xun Wu, Shaohan Huang, Furu Wei. “Learning to Coach for Experiential Learning.” *arXiv preprint*, 2026. [arXiv:2609.15851]
- [2] Tianzhu Ye*, Li Dong*, **Guanheng Chen**, He Zhu, Xun Wu, Shaohan Huang, Furu Wei. “LLM-as-a-Coach: Experiential Learning for Non-Verifiable Tasks.” *arXiv preprint*, 2026. [arXiv:2607.18110]

*Equal contribution.

RESEARCH EXPERIENCE

Microsoft Research Asia (MSRA)

Research Intern, General Artificial Intelligence Group

Mentor: Dr. Li Dong

Mar 2026 – Present

Beijing, China

Learning to Coach for Experiential Learning

- Trains a dedicated coach with reinforcement learning to extract actionable knowledge from an actor’s trajectories, using the frozen actor’s downstream performance as reward.
- Studies same-instance and cross-instance reward objectives to improve subsequent solutions and extract knowledge that transfers across problems.
- Demonstrates iterative test-time improvement on mathematical reasoning and interactive text-games, with transfer to out-of-distribution tasks.

LLM-as-a-Coach: Experiential Learning for Non-Verifiable Tasks

- Uses natural-language experiential knowledge as a richer feedback signal for post-training on open-ended tasks.
- Internalizes coaching feedback through on-policy context distillation, improving generalization beyond rubric-based reinforcement learning.